

**SUPPLEMENTARY MATERIAL FOR
“THE HUMP BETWEEN TWO CLOSE ZEROS: RANDOM UNITARY
MATRICES AND THE RIEMANN ZETA FUNCTION”**

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Numbers without the prefix S (theorems, lemmas, equations, sections, tables) refer to the main text; notation is that of its Section 2.

S1. COMPLEMENTS TO THEOREM 1

Remark S1 (Several angles near the midpoint). The bound (8) does not require the singular factors to be treated one cluster at a time: configurations in which several ϑ_j approach the midpoint simultaneously are included in the exact moment $\mathcal{M}_{N-2}(k+2)$. For a self-contained check of its finiteness, bound the Vandermonde factor of p_{N-2} by $4^{(N-2)(N-3)/2}$; then by Tonelli $\mathbb{E}_{N-2} |\Lambda_{N-2}(1)|^{2a} \leq c_N \prod_{j=1}^{N-2} \int_{\mathbb{T}} |2 \sin(\vartheta_j/2)|^{2a} d\vartheta_j$, a product of one-dimensional integrals, each finite iff $2a > -1$. With $2a = 4 + 2k$ this is $k > -\frac{5}{2}$.

Remark S2 (Uniform integrability). Lemmas B and C give, for $k \in (-\frac{5}{2}, 0]$ and $0 < \varepsilon \leq 1$,

$$\varepsilon^{-3} \mathbb{E} \sum_n \mathbf{1}\{s_n < \varepsilon\} \left(\frac{M_n}{s_n^2} \right)^{2k} \leq \frac{5^{-2k} \mathcal{M}_{N-2}(k+2)}{6\pi},$$

uniformly in ε . Together with Theorem 1(a) this bounds $\sup_{\varepsilon \leq \varepsilon_0} \langle (M/s^2)^{2k} \rangle_\varepsilon$, so by the de la Vallée-Poussin criterion $(M_J/s_J^2)^{2k}$ is uniformly integrable under $\langle \cdot \rangle_\varepsilon$ for every $k > -\frac{5}{2}$ (use an exponent $k' \in (-\frac{5}{2}, k)$). The proof in the main text uses the domination directly.

Remark S3 (Position of the maximum). The maximum of $|\Lambda_N|$ over a small gap sits asymptotically at its midpoint: if t_n^* is the (smallest) maximiser on the n -th gap, then $\langle \mathbf{1}\{|(t^* - \theta_{(\cdot)})/s - \frac{1}{2}| > \eta\} \rangle_\varepsilon \rightarrow 0$ for every $\eta > 0$. Indeed, in the notation of the proof of Theorem 1, with $t = sx$, $x \in [0, 1]$, the normalised profile is $x(1-x) \operatorname{sinc}(\frac{sx}{2}) \operatorname{sinc}(\frac{s(1-x)}{2}) |\Lambda_{N-2}(e^{isx})|$, which converges uniformly in x to $|\Lambda_{N-2}(1)|x(1-x)$; when $\Lambda_{N-2}(1) \neq 0$ the limit has the unique maximiser $\frac{1}{2}$, so the maximisers converge to $\frac{1}{2}$. Lemma C with the bounded integrand $\mathbf{1}\{|x^* - \frac{1}{2}| > \eta\}$, the bound $W_s \leq 4^{2(N-2)}$ and dominated convergence give the claim.

S2. PROOF DETAILS FOR COROLLARY 3

Lemma S1 (Barnes asymptotics). *For fixed real $a > -\frac{1}{2}$, as $n \rightarrow \infty$,*

$$\mathcal{M}_n(a) = \frac{G(1+a)^2}{G(1+2a)} \cdot \frac{G(n+1)G(n+1+2a)}{G(n+1+a)^2} \sim \frac{G(1+a)^2}{G(1+2a)} n^{a^2}.$$

Proof. The identity follows from $\prod_{j=1}^n \Gamma(j+c) = G(n+1+c)/G(1+c)$, a consequence of $G(z+1) = \Gamma(z)G(z)$. The classical expansion $\log G(z+1) = \frac{z^2}{2} \log z - \frac{3}{4}z^2 + \frac{z}{2} \log 2\pi - \frac{1}{12} \log z + \zeta'(-1) + O(z^{-1})$ [5] gives, for fixed c , $\log G(n+1+c) - \log G(n+1) = (cn +$

$\frac{c^2}{2}) \log n - cn + \frac{c}{2} \log 2\pi + o(1)$. The combination with $c = 0, 2a, a, a$ and weights $1, 1, -1, -1$ leaves $(2a^2 - a^2) \log n = a^2 \log n$. This is the asymptotic of [9]. $\square$

Lemma S2 (Digamma sums). *For integers $a \geq 0, n \geq 1$,*

$$(S1) \quad \sum_{j=1}^n \psi'(j+a) = H_{n+a} - H_a - a[\psi'(a+1) - \psi'(n+a+1)] + n\psi'(n+a+1),$$

$$(S2) \quad \sum_{j=1}^{\infty} \psi^{(r-1)}(j+a) = (-1)^r (r-1)! A_a^{(r)} \quad (r \geq 3).$$

Proof. Write $\psi^{(r-1)}(x) = (-1)^r (r-1)! \sum_{m \geq 0} (x+m)^{-r}$ and put $l = j+a+m$. For fixed $l \geq a+1$ the number of $j \in [1, n]$ with $j \leq l-a$ is $\min(n, l-a)$. For $r = 2$ this gives $\sum_{l=a+1}^{a+n} (l-a)l^{-2} + n \sum_{l>a+n} l^{-2}$, which is (S1) since $\sum_{l>c} l^{-2} = \psi'(c+1)$. For $r \geq 3$ and $n = \infty$ it gives $\sum_{l>a} (l-a)l^{-r} = A_a^{(r)}$. All rearrangements are of series of one sign. $\square$

Details for (c). By (10) and (S1) with $n = N-2, \kappa_2 = S_4 - \frac{1}{2}S_2$ where $S_a = \sum_{j \leq n} \psi'(j+a) = \log n + \gamma - H_a - a(\zeta(2) - H_a^{(2)}) + 1 + O(n^{-1})$, using $\psi'(a+1) = \zeta(2) - H_a^{(2)}, n\psi'(n+a+1) = 1 + O(n^{-1})$. With $H_4 = \frac{25}{12}, H_4^{(2)} = \frac{205}{144}, H_2 = \frac{3}{2}, H_2^{(2)} = \frac{5}{4}, \kappa_2 = \frac{1}{2} \log n + \frac{\gamma}{2} + \frac{1}{2} - \frac{25}{12} + \frac{3}{4} - \frac{5}{4} + \frac{205}{36} - 3\zeta(2) + O(n^{-1}) = \frac{1}{2} \log N + \frac{65}{18} + \frac{\gamma}{2} - \frac{\pi^2}{2} + O(N^{-1})$.

Details for (d). For $j = 1, \tau_1 = -2(\zeta(3) - H_4^{(3)}) + \frac{1}{2}(\zeta(3) - H_2^{(3)}) = -0.010261 \dots$. For $j \geq 2$ use $x^{-2} \leq |\psi''(x)| \leq x^{-2} + 2x^{-3}$ (compare $2 \sum_{m \geq 0} (x+m)^{-3}$ with $2 \int_0^\infty (x+t)^{-3} dt$): it suffices that $(j+4)^{-2} > \frac{1}{4}(j+4)(j+2)^{-3}$, i.e. $4^{1/3}(j+2) > j+4$, true for $j \geq 2$. The limit $\sum_{j \geq 1} \tau_j$ is evaluated by (S2). The general r is the same computation; the series converge because $|\psi^{(r-1)}(x)| \leq (r-1)!x^{-r} + (r-2)!x^{1-r}$.

S3. HOW MUCH DO SMALL GAPS CONTRIBUTE TO THE SECOND MOMENT?

Corollary S3. *For $N \geq 2$,*

$$\lim_{\varepsilon \rightarrow 0} \varepsilon^{-7} \mathbb{E} \sum_{n=1}^N \mathbf{1}\{s_n < \varepsilon\} M_n^2 = \frac{\mathcal{M}_{N-2}(3)}{224\pi} = \frac{(N-1)N^2(N+1)^3(N+2)^2(N+3)}{1\,935\,360\,\pi}.$$

Proof. $M_n^2 = \varepsilon^4 (s_n/\varepsilon)^4 (M_n/s_n^2)^2$. Apply (5) with the bounded continuous $F(u, y) = u^4 y^2$: the limit is $\frac{1}{2\pi} \int_0^1 u^6 du \cdot \frac{1}{16} \mathbb{E}_{N-2} |\Lambda_{N-2}(1)|^6 = \mathcal{M}_{N-2}(3)/(224\pi)$; then use (3). $\square$

In unfolded units, $\varepsilon = 2\pi\tilde{\varepsilon}/N$, the contribution of the gaps with $\tilde{s}_n < \tilde{\varepsilon}$ is

$$\mathbb{E} \sum_{\tilde{s}_n < \tilde{\varepsilon}} M_n^2 = \left(\frac{2\pi}{N}\right)^7 \frac{\mathcal{M}_{N-2}(3)}{224\pi} \tilde{\varepsilon}^7 (1 + o(1)), \quad \left(\frac{2\pi}{N}\right)^7 \frac{\mathcal{M}_{N-2}(3)}{224\pi} \sim \frac{\pi^6}{15120} N^2 = 0.0636 N^2.$$

The total second moment. To compare this with the sum over all gaps, let

$$V_N(\vartheta) := \prod_{j=1}^N 2 \sin \frac{\vartheta - \theta_j}{2},$$

a real, real-analytic function with $|V_N(\vartheta)| = |\Lambda_N(e^{i\vartheta})|$ and $V_N(\vartheta + 2\pi) = (-1)^N V_N(\vartheta)$. Winn [16, (1.1)–(1.2)] uses $Z_U(\vartheta) = \prod_j (1 - e^{i(\theta_j - \vartheta)})$ and $V_U(\vartheta) = \exp(iN \frac{\vartheta + \pi}{2} - \frac{i}{2} \sum_j \theta_j) Z_U(\vartheta)$.

Since $1 - e^{i(\theta_j - \vartheta)} = 2i e^{i(\theta_j - \vartheta)/2} \sin \frac{\vartheta - \theta_j}{2}$, we have $Z_U(\vartheta) = i^N e^{\frac{i}{2}(\sum_j \theta_j - N\vartheta)} V_N(\vartheta)$, hence $V_U = e^{iN\pi/2} i^N V_N = (-1)^N V_N$. Following Winn, put $\tilde{F}_N(h, k) := \mathbb{E}|V_N(0)|^{2k-2h} |V'_N(0)|^{2h}$.

Proposition S4 (Total second moment). *Let $N \geq 2$. Almost surely*

$$\sum_{n=1}^N M_n^2 = \int_0^{2\pi} |V_N(\vartheta) V'_N(\vartheta)| d\vartheta,$$

hence $\mathbb{E} \sum_n M_n^2 = 2\pi \tilde{F}_N(\frac{1}{2}, 1)$, and as $N \rightarrow \infty$

$$\mathbb{E} \sum_{n=1}^N M_n^2 \sim \frac{e^2 - 5}{2} N^2 = 1.194528 \dots N^2.$$

Proof. Almost surely the θ_j are distinct. On the n -th gap V_N has no zero, and $(\log |V_N|)' = \frac{1}{2} \sum_j \cot \frac{\vartheta - \theta_j}{2}$ is strictly decreasing (its derivative is $-\frac{1}{4} \sum_j \sin^{-2} \frac{\vartheta - \theta_j}{2}$), with limits $+\infty$ and $-\infty$ at the left and right endpoints. Hence V_N^2 increases up to a unique point t_n^* and decreases after it, vanishing at both endpoints, so that $\int_{\text{gap}} |V_N V'_N| = \frac{1}{2} \int_{\text{gap}} |(V_N^2)'| = V_N(t_n^*)^2 = M_n^2$. The N gaps tile a period of the 2π -periodic function $|V_N V'_N|$, which gives the identity. Rotating the eigenangles by α replaces $V_N(\vartheta)$ by $V_N(\vartheta - \alpha)$, so $\mathbb{E}|V_N(\vartheta) V'_N(\vartheta)|$ does not depend on ϑ and $\mathbb{E} \sum_n M_n^2 = 2\pi \tilde{F}_N(\frac{1}{2}, 1)$. Finally, Winn proved $N^{-2} \tilde{F}_N(\frac{1}{2}, 1) \rightarrow (e^2 - 5)/(4\pi)$ [16, (1.11), proved in §7], a conjecture of Hughes. $\square$

The identity is the random-matrix counterpart of the reduction used by Conrey and Ghosh for Hardy's function (see [16, §1]), and the constants match: under the Riemann Hypothesis $\sum_{0 < \gamma \leq T} \max_{\gamma \leq t \leq \gamma+} |\zeta(\frac{1}{2} + it)|^2 \sim \frac{e^2 - 5}{4\pi} T (\log T)^2$ [6], i.e. a mean of $\frac{e^2 - 5}{2} \log T$ per gap, against $\frac{e^2 - 5}{2} N$ per gap here. Winn's Theorem 2.2 [16] also evaluates $\tilde{F}_N(\frac{1}{2}, 1)$ at finite N ; for $k = 1$, $h = \frac{1}{2}$ it reduces to (see below)

$$(S3) \quad \mathbb{E} \sum_{n=1}^N M_n^2 = 2(N+1) \left[N - \sum_{p=2}^N \frac{(p-2)! 2^p N!}{(N-p)! (p+1)! p! N^{p-1}} \right]$$

(for $N = 2$ this is 10, as a direct computation confirms), so that $\mathbb{E} \sum_n M_n^2 / N^2 = 1.6055, 1.4381, 1.3457, 1.2949, 1.2696$ for $N = 6, 10, 16, 24, 32$. A Monte Carlo run (4×10^4 matrices for $N \leq 16$, 2×10^4 for $N = 24, 32$; maxima on the 63 interior points of a 65-point grid per gap with parabolic refinement) gives 1.604(5), 1.455(7), 1.339(8), 1.284(14), 1.258(14); the $N = 10$ value is 2.6σ high, and an independent run with 4×10^5 matrices gives 1.4394(20) there.

Gaps below $\tilde{\varepsilon}$ therefore carry a fraction $c_N \tilde{\varepsilon}^7 (1 + o(1))$ of the second moment, with $c_N = 0.140$ ($N = 6$) decreasing to 0.066 ($N = 32$) and $c_N \rightarrow \pi^6 / (7560(e^2 - 5)) = 0.0532$; this is about 10^{-6} for $\tilde{\varepsilon} = 0.2$. Small gaps are invisible in the mean of M_n^2 , which is the quantity controlled in [6, 8].

Winn's formula for $h = \frac{1}{2}$, $k = 1$. We specialise [16, Thm. 2.2, eq. (2.9)] to $h = \frac{1}{2}$, $k = 1$, in Winn's notation. For a partition λ and $b \in \mathbb{R}$, $[b]_\lambda = \prod_{i=1}^{\ell(\lambda)} (b - i + 1)_{\lambda_i}$ with the rising factorial $(x)_m = x(x+1) \cdots (x+m-1)$, h_λ is the product of the hook lengths of λ , and

$$C_N(p, k) = (-2)^p \sum_{\lambda \vdash_{kp} N} \frac{[k]_\lambda [-N]_\lambda}{[2k]_\lambda h_\lambda^2},$$

the sum running over partitions of p into at most k parts [16, (2.1)–(2.5)]. For half-integer h and $k \in \mathbb{N}$, [16, (2.9)] reads

$$\begin{aligned} \tilde{F}_N(h, k) = \frac{2(-1)^{h+1/2}}{2^{2h}\pi} \tilde{F}_N(0, k) & \left\{ \sum_{p=1}^{2h} \sum_{\ell=1}^p \binom{2h}{p-\ell} \frac{(-1)^\ell}{\ell} (-N)^{2h-p} p! C_N(p, k) \right. \\ & \left. + \sum_{p=2h+1}^{kN} \frac{(2h)!(p-2h-1)!}{N^{p-2h}} C_N(p, k) \right\}. \end{aligned}$$

Step 1: the ingredients. For $h = \frac{1}{2}$, $k = 1$: the prefactor is $2(-1)^1/(2\pi) = -1/\pi$; $\tilde{F}_N(0, 1) = \mathcal{M}_N(1) = \prod_{j=1}^N \frac{\Gamma(j)\Gamma(j+2)}{\Gamma(j+1)^2} = \prod_{j=1}^N \frac{j+1}{j} = N+1$ by (2); the first double sum has the single term $p = \ell = 1$, equal to $\binom{1}{0} \frac{(-1)^1}{1} (-N)^0 1! C_N(1, 1) = -C_N(1, 1)$; and the second sum runs over $2 \leq p \leq N$ with weight $(p-2)!/N^{p-1}$.

Step 2: the partition sum. For $k = 1$ only the one-part partition $\lambda = (p)$ contributes. Its hook lengths are $p, p-1, \dots, 1$, so $h_{(p)} = p!$, and $[b]_{(p)} = (b)_p$, so that $[1]_{(p)} = p!$, $[2]_{(p)} = (p+1)!$ and $[-N]_{(p)} = (-N)(-N+1) \cdots (-N+p-1) = (-1)^p N!/(N-p)!$ for $p \leq N$. Hence

$$C_N(p, 1) = (-2)^p \frac{p! (-1)^p N!/(N-p)!}{(p+1)! (p!)^2} = \frac{2^p N!}{(N-p)! (p+1)! p!}, \quad C_N(1, 1) = N.$$

Step 3: assembly. Collecting,

$$\tilde{F}_N(\tfrac{1}{2}, 1) = \frac{N+1}{\pi} \left[N - \sum_{p=2}^N \frac{(p-2)! 2^p N!}{(N-p)! (p+1)! p! N^{p-1}} \right],$$

and $\mathbb{E} \sum_n M_n^2 = 2\pi \tilde{F}_N(\tfrac{1}{2}, 1)$ (Proposition S4) gives (S3). Each term is rational, so $\mathbb{E} \sum_n M_n^2 \in \mathbb{Q}$; for $N = 2, 3, 4, 6$ it equals $10, \frac{496}{27}, \frac{233}{8}, \frac{42134}{729}$.

Checks. (i) For $N = 2$ the bracket is $2 - \frac{1}{3}$, so the formula gives 10. Directly: let $\phi \in (0, 2\pi)$ be $\theta_2 - \theta_1$ modulo 2π ; the two gaps are ϕ and $2\pi - \phi$, and by the proof of Lemma A (with $N - 2 = 0$ other angles) a gap of length s has hump $4 \sin^2 \frac{s}{4}$, attained at its midpoint. The humps are thus $4 \sin^2 \frac{\phi}{4}$ and $4 \cos^2 \frac{\phi}{4}$, so $\sum_n M_n^2 = 16(\sin^4 \frac{\phi}{4} + \cos^4 \frac{\phi}{4}) = 16 - 8 \sin^2 \frac{\phi}{2}$, and ϕ has density $\sin^2(\phi/2)/\pi$ on $(0, 2\pi)$ by (1), so $\mathbb{E} \sin^2 \frac{\phi}{2} = \frac{3}{4}$ and $\mathbb{E} \sum_n M_n^2 = 10$. (ii) As $N \rightarrow \infty$, $N!/((N-p)!N^p) \rightarrow 1$ for each p and the terms are dominated by $2^p(p-2)!/((p+1)!p!)$, so $N^{-2} \tilde{F}_N(\tfrac{1}{2}, 1) \rightarrow \frac{1}{\pi} \left[1 - \sum_{p \geq 2} \frac{2^p(p-2)!}{p!(p+1)!} \right] = \frac{e^2-5}{4\pi}$, which is [16, (7.8)].

S4. MONTE CARLO VERIFICATION IN FULL

Method. Haar unitaries were generated by the QR decomposition of complex Ginibre matrices with the phases of the diagonal of R removed [12]. For every gap with unfolded length $\tilde{s} < 0.4$, $\log |\Lambda_N|$ was evaluated on the 39 interior points of a uniform 41-point grid across the gap and the maximum refined by a parabola through the largest grid value and its neighbours. Statistics are reported in the bands $\tilde{s} < \tilde{\varepsilon}$ for $\tilde{\varepsilon} \in \{0.4, 0.3, 0.2, 0.1, 0.05\}$, in angle units (M/s^2); the predictions are Theorem 1 with Theorem 2. The “rate” is $\#\{\text{gaps} < \varepsilon\}/(\text{matrices} \cdot \varepsilon^3)$, to be compared with $N^2(N^2 - 1)/(72\pi)$. No zeta data enter this section.

N	matrices	events	$\mathbb{E}[M/s^2]$			rate		
			Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$
6	$6 \cdot 10^5$	3 724	1.4784	1.4828(30)	1.4751(83)	5.570	5.392	5.405
10	$4 \cdot 10^5$	4 302	3.9279	3.935(11)	3.919(31)	43.77	42.96	43.36
16	$3 \cdot 10^5$	5 253	10.194	10.178(35)	10.059(98)	288.6	280.4	289.1
24	$2 \cdot 10^5$	5 229	23.903	24.13(10)	24.10(29)	1464	1419	1457

N	$\mathbb{E} \log(M/s^2)$			$\text{Var} \log(M/s^2)$			κ_3
	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm	$\tilde{s} < 0.2$	$\tilde{s} < 0.1$	Thm
6	0.3232	0.3276(22)	0.3229(62)	0.1486	0.1447	0.1448	-0.0439
10	1.2292	1.2323(29)	1.2330(81)	0.2987	0.2963	0.2834	-0.0688
16	2.1007	2.0998(34)	2.0822(95)	0.4691	0.4664	0.4781	-0.0872
24	2.8723	2.8827(39)	2.8804(110)	0.6340	0.6306	0.6326	-0.0990

TABLE S1. CUE Monte Carlo against Theorems 1 and 2 (angle units; statistical errors in the last digits; “events” is the number of gaps with $\tilde{s} < 0.1$). The lower half is Table 1. Files `200t_mc_N{6,10,16,24}.json`.

Results. Table S1 shows agreement with the theorems at the level of the statistical errors: at $\tilde{s} < 0.1$ every deviation of the mean hump and of the log-mean is below 2σ and the counts agree within 2σ (Poisson errors). At $\tilde{s} < 0.2$ the small finite- ε bias discussed below is visible (largest in the log-mean at $N = 24$, $+0.010$). Further checks:

- *Independence.* The correlation of $\tilde{s}$ with $\log(M/s^2)$ falls with the band: 0.021, 0.013, 0.007, 0.000 at $\tilde{\varepsilon} = 0.4, 0.3, 0.2, 0.1$ ($N = 10$, file `200t_mc_N10.json`).
- *Lemma A.* The largest difference between $\log(M/s^2)$ and $\log(|\Lambda_{N-2}(\text{mid})|/4)$ over all events with $\tilde{s} < 0.1$ is at most 0.042 (all N ; files of Table S1).
- *Third cumulant.* Independent runs with 8×10^5 ($N = 10$) and 4×10^5 ($N = 16$) matrices give $\kappa_3 = -0.0639$ (8 526 events) and -0.0863 (7 021 events) at $\tilde{s} < 0.1$, against -0.0688 and -0.0872 (files `200t_mc3_N{10,16}.json`); the same runs give $\mathbb{E} \log(M/s^2) = 1.2357(59)$ and $2.1021(81)$.
- *Theorem 2 directly.* Sampling the ξ_j by rejection (4×10^5 products) gives, for $N = 10$, $\mathbb{E} \log(X/4) = 1.2286$ (1.2292), $\text{Var} = 0.2976$ (0.2987), $\kappa_3 = -0.0675$ (-0.0688), $\mathbb{E} X/4 = 3.924$ (3.928); at $N = 24$ the mean, variance and $\mathbb{E} X/4$ agree within 0.2%, and $\kappa_3 = -0.0963$ (-0.0990) (script `200t_carpim.py`, output `not8_ek/200t_carpim_seed7.out`).
- *Finite ε .* In a run with 3×10^6 matrices at $N = 10$ (file `not8_ek/200t_mc_N10_3e6_seed7.json`) the excess of $\mathbb{E}[M/s^2]$ over its limit is 0.0771(15), 0.0440(23), 0.0198(41) at $\tilde{\varepsilon} = 0.4, 0.3, 0.2$: the successive ratios 1.75 and 2.22 match $(4/3)^2 = 1.78$ and $(3/2)^2 = 2.25$, as expected from Remark 1. At $\tilde{\varepsilon} = 0.2$ the excesses of $\mathbb{E} \log$, $\text{Var} \log$ and κ_3 are $+0.0070(11)$, -0.005 and $+0.004$.

S5. THE ZETA SIDE BEFORE THE TESTS

Weak points of Conjecture 4. Recorded before the tests: (W1) the independence of P_X from the conditioning, whereas the prime waves modulate the local density of zeros at relative order $1/L$ (Landau’s formula [10, 7]) and imprint on the gaps, as measured in the

companion notes [2, 3] — and at $b = 2$ the arithmetic term is not a small correction (κ_3 : CUE baseline -0.069 , arithmetic shift $+0.234$ at $L \approx 10$); (W2) finite L , where [1] found effective dimensions $N = L + c$; (W3) the order of limits and the transfer of finite- ε corrections; (W4) the passage from (ii) to (iii), which interchanges $L \rightarrow \infty$ with differentiation in k at $k = 0$.

Finite L . At $L \approx 11$ the form $a_k \times$ (tilted CUE) is not the moment sequence of a probability law: $\log |a_{i\omega/2}|$ grows faster than ω^2 (each prime $p \lesssim \omega^2$ contributes), while the characteristic function of the tilted CUE decays only like $e^{-\omega^2 \log N/4}$. Its low cumulants are nevertheless defined. Genuine probability models are the hybrids $\text{hyb}(X)$: an independent random Euler product over $p \leq X$ times a tilted CUE of dimension $N_X = L/(e^\gamma \log X)$. Table S2 lists the predictions at $L = 10$. At $b = 2$ every arithmetic model predicts a positive κ_3 , whereas the pure tilted CUE gives -0.069 and a stationary Gaussian surface with the same power spectrum gives $\kappa_3 = \psi''(3/2)/8 = -0.1036$ and $\kappa_2 = \psi'(3/2)/4 = 0.2337$, independent of L (the two-point Kac–Rice density makes $|X''|/\sigma$ a χ_3 variable). The pre-test headline was therefore a sign: *close-pair log-humps of zeta are skewed to the right*, with skewness $\kappa_3/\kappa_2^{3/2} \approx +1.7$ at $L = 10$ under a_k .

b	CUE	a_k	hyb $X=2$	$X=3$	$X=5$	$X=7$
0	1.940/−2.393	1.852/−2.159	2.126/−2.233	2.080/−2.126	1.996/−2.042	1.977/−1.991
1	0.568/−0.194	0.480/+0.039	0.775/−0.039	0.790/+0.050	0.778/+0.103	0.803/+0.131
2	0.299/−0.069	0.211/+0.165	0.523/+0.083	0.585/+0.160	0.622/+0.197	0.674/+0.213

TABLE S2. Pre-test predictions κ_2/κ_3 of the log-observable on rung b at $L = 10$. Gaussian surface: $b = 0$: $\pi^2/8 = 1.234$, $-7\zeta(3)/4 = -2.104$; $b = 1$: $\pi^2/24 = 0.411$, $-\zeta(3)/4 = -0.300$; $b = 2$: 0.234 , -0.104 .

S6. THE PRE-REGISTERED TESTS IN FULL

Protocol. Each test was frozen — predictions, thresholds, analysis code — in a pre-registration file that was committed with its sha256 stamp to the author’s private working repository, and pushed, before the measurement: 200-A on 26 September 2026, 17:12 (UTC+3), 200-B on 26 September 2026, 20:19, 200-C on 27 September 2026, 16:22. Before each measurement no distribution summary (moment, cumulant, quantile, histogram) of the tested observable had been computed on the data used; engine checks reported only differences between two engines. (The only earlier look at a rung-0 distribution in the project was a Kolmogorov–Smirnov test of $\log |Z(t)|$ against Selberg’s normal law at low height; Z' at the zeros and the humps of close pairs had never been computed.) Test 200-B was designed after 200-A was sealed, and 200-C after 200-B. Test 201 (phase lock) was frozen in the same way on 28 September 2026, 09:34, and committed and pushed to the (private) repository before its measurement; no phase statistic of the zeros had been computed before (only synthetic data).

Data and engine. Tests 200-A/B use A. M. Odlyzko’s first 2×10^6 zeros [14, 13] in three windows: W1, zeros 20 868–198 239 ($L \in [8, 10)$, $\bar{L} \approx 9.3$); W2, 198 239–598 742 ($[10, 11)$, 10.6); W3, 598 742–2 001 052 ($[11, 12.1]$, 11.65). Test 200-C uses four LMFDB files of zeros computed by Platt [11, 15], 1.5×10^6 consecutive zeros each, at $\bar{L} = 14.19, 16.58, 18.88, 22.31$ (windows C1–C4). Z and Z' are computed by a vectorised Riemann–Siegel formula

with the C_0 and C_1 remainder terms (anchored phases in C1–C4), checked against `mpmath` ($|\Delta Z| \leq 2.4 \times 10^{-7}$ in W1–W3, $\sim 10^{-9}$ in C1–C4). On rung $b = 0$ we use 10^6 uniform random t per window; on $b = 1$ all zeros (in C4, 10^6 random zeros); on $b = 2$ all pairs with $\tilde{\delta}_n < \varepsilon$, primary $\varepsilon = 0.2$ (1 143, 2 860, 10 165 pairs in W1–W3; 11 515 to 12 143 in each of C1–C4), with M_n from a 129-point grid and parabolic refinement. Statistics are unbiased k -statistics k_2, k_3 ; errors are jackknife errors over 64 contiguous t -blocks (128 where a pre-registered block-sensitivity rule required it), rescaled by factors f calibrated on synthetic data processed by the same pipeline ($f = 0.82$ – 1.22 in 200-A, 0.94 – 1.13 in 200-B, 0.94 – 1.25 in 200-C).

Shifts. The *arithmetic shift* of rung b is $s_b := k_3 - \kappa_3^{\text{CUE},b}(\bar{L})$ (similarly for k_2). On $b = 2$ we also subtract the finite- ε correction $\Delta\kappa_r^{\text{CUE}}(\varepsilon)$ from a CUE Monte Carlo at the same ε and $N \approx \bar{L}$, and add 100% of it to the error, since its transfer to zeta is assumed, not tested; this defines $D_r := k_r - \kappa_r^{\text{CUE},2}(\bar{L}) - \Delta\kappa_r^{\text{CUE}}(\varepsilon)$, pooled over windows by inverse variance ($\bar{D}_r$).

Test 200-A: the lower rungs. A family of parameter-free models — CUE, a_k , $\text{hyb}(X)$ for $X \in \{2, 3, 5, 7, 11\}$, Gaussian — was compared on the twelve numbers (k_2, k_3 on two rungs in three windows) with the full jackknife covariance plus a pre-declared theory margin (0.0188 for κ_2 , 0.0334 for κ_3).

Observation S1 (Lower rungs; pre-registered). *The arithmetic signature is present ($\chi_{\text{CUE}}^2 - \chi_{a_k}^2 = 287.1$, threshold 25), and the selected model is a_k , decisively (runner-up CUE at $\Delta\chi^2 = 287.1$, threshold 9), with goodness of fit $\chi_{\min}^2/12 = 28.2/12 = 2.35$ (“good”). The two rungs separately select a_k ($\chi^2 = 6.1$ and 22.1). The hybrids and the Gaussian surface are rejected (χ^2 : CUE 315.2, a_k 28.2, hybrids 872–1243, Gaussian 3689.7; a continuous- X hybrid is best at $X = 2.99$ with $\chi^2 = 129.3$).*

Post hoc, the shifts tell a more detailed story (Table S3). The variance shift is nearly rung-independent and close to κ_2^{arit} : -0.0820 ± 0.0024 at $b = 0$ and -0.0726 ± 0.0008 at $b = 1$, against -0.0881 . The skewness shift is not: $+0.283 \pm 0.009$ at $b = 0$ and $+0.150 \pm 0.001$ at $b = 1$, against $+0.2337$, stable across the three windows. A ladder with the same arithmetic contribution on every rung therefore fails at the level of κ_3 at these heights. Before 200-B we registered, besides a_k , two extrapolations of the two lower rungs (not theories): geometric decay, $s_2 = s_1^2/s_0$, predicting $\bar{D}_3 = +0.0794 \pm 0.0028$; and linear decay, $s_2 = 2s_1 - s_0$, predicting $+0.017 \pm 0.0095$ (grouped with pure RMT, $\bar{D}_3 = 0$, as “extinguished”).

Test 200-B: close pairs.

Observation S2 (Close pairs; pre-registered, blind). *With 14 168 close pairs ($\tilde{\delta} < 0.2$, $L \approx 9$ – 12):*

- (1) *The arithmetic signature is present on $b = 2$: $\bar{D}_3 = +0.0859 \pm 0.0030$ (29σ).*
- (2) *The sign of the skewness flips: the pooled corrected $\kappa_3 = +0.0143 \pm 0.0030$ (4.8σ). The skewness $\kappa_3/\kappa_2^{3/2}$ is about $+0.15$, not the $+1.7$ of the a_k form: the headline prediction held in sign, not in size.*
- (3) *The a_k model carried over from 200-A holds for the variance ($\bar{D}_2 = -0.0988$ against -0.0881 , tolerance 0.024) and fails for the third cumulant (difference -0.148 , tolerance 0.142): partial.*

- (4) *The shape of the ladder is geometric, decisively: $\chi^2 = 0.36$ (geometric), 24.0 (extinguished), 200.7 (constant a_k). The blind geometric prediction $+0.0794 \pm 0.0028$ differs from the measurement by 0.0065 (1.6σ).*

Secondary results (recorded, not decisive): $\bar{D}_3 = +0.0867 \pm 0.0066$ at $\varepsilon = 0.1$ and $+0.0828 \pm 0.0044$ at $\varepsilon = 0.3$; per window $+0.076(6)$, $+0.084(5)$, $+0.093(5)$, a mild rise with height ($\approx 2.5\sigma$). The variance shift on $b = 2$ is -0.0988 ± 0.0036 , far from the geometric and linear extrapolations (-0.064 , $\approx 9\sigma$). Theorem 1(b) has zeta counterparts: the Pearson correlation of $\tilde{\delta}_n$ and $\log Y_n$ is $+0.021$, -0.004 , -0.005 in W1–W3 (CUE Monte Carlo: $+0.005$ to $+0.008$), and the fraction of events with $\tilde{\delta}_n < \varepsilon/2$ is 0.124 – 0.129 (the $U^{1/3}$ law gives 0.125 , the CUE at finite ε 0.128).

rung	observable	κ_3 shift	ratio to previous rung	κ_2 shift
$b = 0$	$\log Z(t) $, random t	$+0.2828 \pm 0.0093$	—	-0.0820 ± 0.0024
$b = 1$	$\log Z'(\gamma_n) $	$+0.1499 \pm 0.0009$	0.530 ± 0.018	-0.0726 ± 0.0008
$b = 2$	$\log(M_n/\tilde{\delta}_n^2)$, $\tilde{\delta}_n < 0.2$	$+0.0859 \pm 0.0030$	0.573 ± 0.020	-0.0988 ± 0.0036
a_k	(every rung)	$+0.2337$	1	-0.0881

TABLE S3. Arithmetic shifts (observed minus tilted CUE at $N = \bar{L}$), pooled over W1–W3 ($L \approx 9$ – 12).

Test 200-C: height. Is the decay a property of the ladder or of the height? Before 200-C, a post hoc fit of the A/B window values to $s_b(L) = a_k + c_b L^{-\nu}$ gave $\nu = 0.48 \pm 0.12$; three hypotheses were registered for the new windows: H_C (shifts constant at their A/B values), $H_{S\nu}$ (the fitted relaxation towards a_k) and H_{S1} ($\nu = 1$). The exponent is called γ in the pre-registration files.

Observation S3 (Height; pre-registered, blind). *On C1–C4 ($L = 14.2$ – 22.3), the class is $H_{S\nu}$, decisively ($\chi^2 = 10.3$ for 12 points; H_{S1} 30.1, H_C 166.2). Fitted on the new windows alone with asymptote a_k , the exponent is $\hat{\nu} = 0.749 \pm 0.170$ (4.4σ from zero: finite height). Over all seven windows with a free common asymptote, $\hat{A} = 0.277 \pm 0.041$ (within 1.0σ of $\kappa_3^{\text{arit}} = 0.2337$) and $\hat{\nu} = 0.447 \pm 0.144$.*

The per-window shifts are in Table 2. The blind $H_{S\nu}$ predictions for C1–C4 were: $b = 1$: $+0.159$, $+0.164$, $+0.169$, $+0.174$; $b = 2$: $+0.105$, $+0.114$, $+0.121$, $+0.130$. The observed $b = 1$ shifts lie 0.005 – 0.01 above them (faster relaxation). The variance shifts do not depend on height: -0.082 to -0.094 ($b = 0$), -0.075 to -0.079 ($b = 1$), -0.095 to -0.115 ($b = 2$), around $\kappa_2^{\text{arit}} = -0.088$. The gap–hump correlation agrees with the CUE Monte Carlo ($|\rho| \leq 0.01$).

Test 201: phase lock at the conditioned points. The ladder concerns the size of $|Z|$ at conditioned points; a simpler question is where the conditioned points sit relative to the phase of a prime wave $\cos(t \log p)$. For single zeros the answer is Landau’s formula [10] (uniform version in [7]): for fixed $x > 1$, $\sum_{0 < \gamma \leq T} x^\rho = -\frac{T}{2\pi} \Lambda(x) + O(\log T)$, which for zeros on the critical line and a window at height t means

$$\mathbb{E}[\cos(m\gamma_n \log p)] \approx -\frac{\log p}{p^{m/2}L}, \quad \mathbb{E}[\sin(m\gamma_n \log p)] \approx 0.$$

Heuristically, the explicit formula modulates the local density of zeros, $n(t) \approx \frac{L}{2\pi} [1 - \frac{2}{L} \sum_{p,m} \frac{\log p}{p^{m/2}} \cos(mt \log p)]$, and the zeros avoid the crests of the prime waves; averaging $\cos(mt \log p)$ against $n(t)$ gives the same values. For the midpoints of close pairs we registered three classes for the ratio R_2 of their lock to the single-zero value: 1 (a pair behaves like one zero), 2 (the two densities multiply) and 4. The last is a heuristic, not a theorem: if the zeros are locally GUE at density $n(t)$, the density of pairs with gap δ at midpoint t is $n(t)^2 [1 - \text{sinc}^2(\pi \delta n(t))] \approx \frac{\pi^2}{3} \delta^2 n(t)^4$, and since the pairs are selected by $\tilde{\delta}_n = \delta_n L / 2\pi < \varepsilon$, i.e. with the mean rather than the local density, their midpoints are weighted by $n(t)^4$ and lock four times as strongly as single zeros; for larger ε the weight tends to $n(t)^2$, and R_2 towards 2.

On the seven windows W1–W3, C1–C4 and the primes $p \leq 13$ (ratios pooled over primes with the jackknife covariance, then over windows):

Observation S4 (Phase lock; pre-registered). (1) *Single zeros follow Landau’s formula to about four decimals: the ratio of the measured lock to $-\log p / (\sqrt{p} L)$ is $R_1 = 1.0000 \pm 0.00004$ (per window from 1.003 ± 0.007 in W1 to 1.00003 ± 0.00009 in C4), also at the second harmonic (1.0000 ± 0.0001), with no drift in $\log L$; at C4, for $p = 2$, the mean of $\cos(\gamma_n \log 2)$ is -0.021975 against -0.021973 . At random t the same statistic is -0.010 to $+0.005$ (control).*

(2) *Midpoints of close pairs ($\tilde{\delta}_n < 0.2$) lock $R_2 = 3.874 \pm 0.023$ times as strongly: class 4, with the classes 1 and 2 excluded. The ratio decreases with ε , $R_2 = 4.07 \pm 0.06$, 3.87 ± 0.02 , 3.74 ± 0.01 at $\varepsilon = 0.1, 0.2, 0.3$, as the heuristic expects; at the second harmonic it is 3.06 ± 0.05 , consistent with the second-order terms of the $n(t)^4$ weight (a synthetic process with that weight gives 2.90).*

For single zeros this verifies a theorem rather than discovers anything; a by-product is that the block-to-block fluctuations of these averages are 7–19 times smaller than for independent points, a sign of the rigidity of the zeros. The lock of close pairs has the size the local-GUE heuristic predicts, so there is no excess lock. It therefore does not by itself explain the decay of the κ_3 shift along the ladder: a translation of the measured lock into cumulant shifts through independent Euler factors was inconclusive (its sign depends on the harmonic cut-off), and the mechanism of the decay remains open. A phase bias of the primes at the conditioned points, which weakens like $1/L$, would produce a relaxation of the kind seen in 200-C; as in [4], where a post hoc pattern $\eta_p \approx p^{-1/3}$ of transfer factors turned out to be a finite-height effect, a pattern seen at one height (≈ 0.55 per rung) was a coincidence of height.

S7. NUMERICAL DETAILS OF THE RATIOS-CONJECTURE COMPUTATIONS

Rung $b = 0$, κ_3 . We use primes up to 10^6 with integral tails, Gauss–Legendre panels adapted to the scales x_1 and $1/L$ (refining from 16 to 24 nodes changes κ_3 by $6 \cdot 10^{-4}$), and drop the second swap term for $w > 0.75$, where it carries $e^{-0.75L}$ (changing the cut to 0.5 moves κ_3 by $6 \cdot 10^{-5}$).

Rung $b = 1$, variance. Near $a = 0$ terms of order $1/a$ cancel; we integrate from $a = 0.01/L$, adding the remaining strip linearly (the inner integral vanishes linearly there). For ζ , 12, 16 and 20 nodes per panel agree to $5 \cdot 10^{-6}$, and the swap cut changes the result by at most $6 \cdot 10^{-4}$ (at $L = 9.35$).

Rung $b = 1$, third cumulant. In $\kappa_3 = \kappa_3^{\text{CUE}}(L) + \iiint (I_\zeta - I_{\text{CUE}})$ the difference of the integrands is integrated over shifts ≥ 0.0075 , with the thin strips below added linearly (cuts

from 0.005 to 0.01 change κ_3 by $5 \cdot 10^{-5}$, the grid by 10^{-4}). The interpolation of A_{dbl} outside its region of convergence moves κ_3 by at most $3 \cdot 10^{-4}$.

S8. REPRODUCIBILITY: FILE GUIDE

All code, data products, pre-registration files (with sha256 stamps and dates) and the research log, including the failed predictions, are available in the repository <https://github.com/azizran/riemann-zero-statistics> (directory `qm_riemann/`), archived at Zenodo, doi:10.5281/zenodo.22942475. The commit hashes below refer to the author's private working repository, in which the pre-registrations and predictions were committed before the measurements; its history can be shown to referees on request. Paths below are relative to `qm_riemann/200_configs/`, whose subdirectory `not8_ek/` holds the runs added for this note.

- Tables 1 and S1, independence and Lemma A checks: `200t_mc_N*.json` ($N = 6, 10, 16, 24$; script `200t_mc.py`, predictions `200t_tahmin.py`); third cumulant: `200t_mc3_N*.json` ($N = 10, 16$); finite ε : `not8_ek/200t_mc_N10_3e6_seed7.json` (arguments 10 3000000 7); Theorem 2 by sampling and κ_2^{arit} : `200t_carpim.py`, output `not8_ek/200t_carpim_seed7.out`.
- Proposition S4: Monte Carlo `not8_ek/m2total.py`, output `not8_ek/m2total_seed11.out`; independent run `not8_ek/m2total_N10_4e5_seed2026.json`; exact values from Winn's formula `not8_ek/m2_exact_winn.json` (these two JSON files contain their generating scripts).
- Constants, ladder baselines and Table S2: `200t_sabitler.py`, `200t_merdiven.py`, `200t_hibrit.py`, `200t_aritmetik_kesin.py`.
- Tests 200-A/B/C: pre-registrations `ONKAYIT_200A/B/C.json` (sha256 prefixes `f0f0ce57`, `b37dc244`, `391257a4`; commits `aa861b5`, `85f0489`, `567af03`), verdicts `HUKUM_200A/B/C.json` (which also hold the CUE Monte Carlo comparisons of Section S6), synthetic calibrations and finite- ε CUE corrections `M6_200A.json`, `M6b_200B.json`, `M6c_200C.json`. Test 201: directory `qm_riemann/201_configs/`, pre-registration `ONKAYIT_201.json` (sha256 prefix `ecc7cf75`, commit `16e7c3b`), verdict `HUKUM_201.json`.
- Table 3, Section 9 and Appendix A: directory `qm_riemann/203_configs/` (`203_oran_k2.py`, `203_oran_k3.py`, `203_oran_b1.py`, `203_b1k3_zeta.py` with its CUE and per-prime checks `203_cue4.py`, `203_b1k3_cue.py`, `203_yerel4.py`); predictions committed before comparison (`182190f`, `bf214f0`, `a056eb0`, `e5f2580`), post-audit correction `62b36dd`, the test against zeros in `sifir_sinamasi_b1k3/`, Figure 1: `figures/ladder_ratios.py`; report `203_ORAN_KAPPA_RAPOR.md`.

The LMFDB zero files are not redistributed; their sha256 stamps are in `ONKAYIT_200C.json`.

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